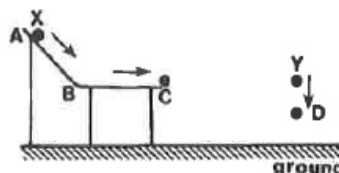


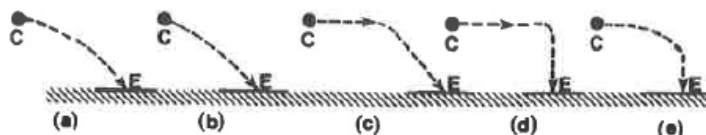
(VI) The left side of the accompanying figure shows a ball X in motion. The ball slides down an incline AB, then on a *frictionless, horizontal* track BC. At C, the ball leaves the track. Ignoring air resistance:



(13) The speed of the ball on the horizontal track BC is:

- (a) Constant.
- (b) Continuously increasing.
- (c) Continuously decreasing.
- (d) Increasing for a while, and constant thereafter.
- (e) Constant for a while, and decreasing thereafter.

(14) Which of the paths below will the ball follow *after* it leaves the track at C?



(15) The *speed* of the ball along the path you have chosen is:

- (a) Constant.
- (b) Continuously increasing.
- (c) Continuously decreasing.
- (d) Constant for a while, and increasing thereafter.
- (e) None of the above.

(16) Beyond C, and *along the path* you have chosen in question (14), what *force(s)* act on the ball?

- (a) The weight of the ball, vertically downward.
- (b) A horizontal force that maintains the motion.
- (c) A force whose direction changes in the direction of motion.
- (d) The weight of the ball and a horizontal force.
- (e) The weight of the ball and a force in the direction of motion.

At the *same time* that ball X in the original figure of part VI above leaves the track at C, and from the *same height* as C, a ball Y identical to ball X is released from rest to fall vertically downward.

(17) When ball Y reaches *point D*, where would ball X be?

- (a) At the same height as D.
- (b) Above D.
- (c) Below D.
- (d) The position of X would depend on how high D is.
- (e) None of the above.

(18) Which ball reaches the ground first?

- (a) Ball X.
- (b) Ball Y.
- (c) The two balls reach ground at the same time.
- (d) It depends on how high C is.
- (e) None of the above.

(19) Which ball reaches the ground with the *greatest speed*?

- (a) Ball X.
- (b) Ball Y.
- (c) The two balls reach ground at the same speed.
- (d) It depends on how high C is.
- (e) None of the above.

(20) If track BC was extended beyond C so that ball X would never leave that track, where would that ball be at the *same time* ball Y hits the ground?

- (a) Vertically above E, where ball X hit the ground in question (14).
- (b) To the right of E.
- (c) To the left of E.
- (d) It depends on how high C is.
- (e) None of the above.